

Erman Patterson

HW 1 Q1

$$7 = 123456789 = n^{\frac{1}{\log n}} = \left(\frac{1}{2}\right)^n$$

$$\log \log n$$

$$\log n = \log(n^3)$$

$$(\log n)^2$$

$$\log_4 n$$

$$n^{-2}$$

$$\sqrt{n} = n^{\frac{1}{2}}$$

$$n^{\frac{3}{4}}$$

$$n = n^{0.5} = n^{1/2}$$

$$n \log n$$

$$n^{\frac{4}{3}}$$

$3^{\log n}$

$$n^2 = n^2 \cdot 10^{100} n \log n = 4^{100n}$$

$$n^2 \log n$$

$$n^4$$

$$2^n$$

$$n 2^n$$

$$4^n = 4^{n-1}$$

$$4^{2n}$$

$$4^{n^2}$$

$$n^n$$

$$n!$$